Git and Github Installation Guide:

Git is installed using different methods depending on your operating system. First, find the guide to install Git on your operating system below, then see the GitHub setup section at the end of the document.

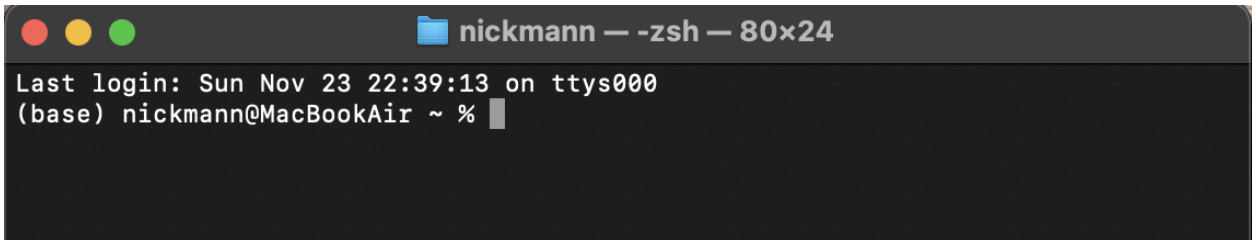
macOS - Installing Git:

We will install Git on your computer using homebrew, but we will need to download it first, if you don't have it (you wouldn't unless you've downloaded previously)

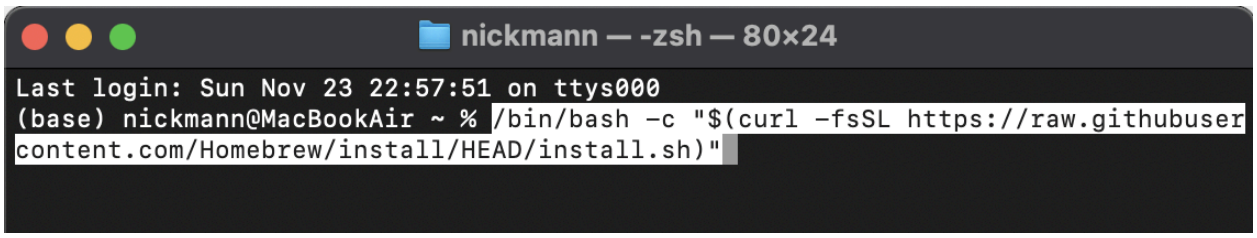
Click on the magnifying glass in the top right corner of your screen to use spotlight search, then type in "terminal". You'll want to click the app highlighted in the image below.



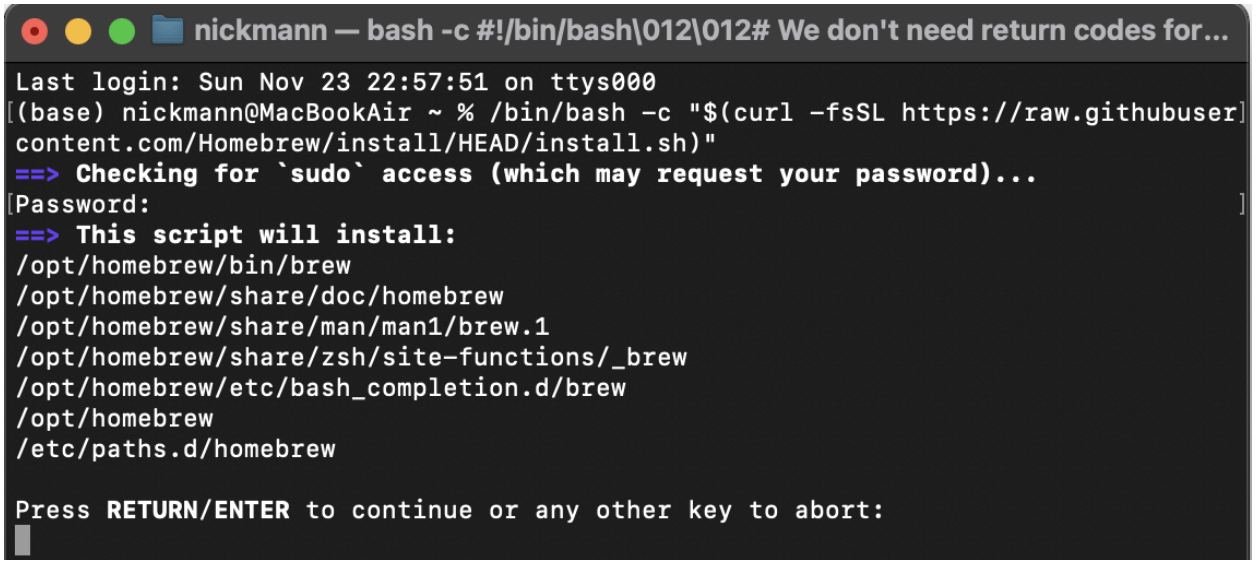
Upon clicking this, a window should pop up:



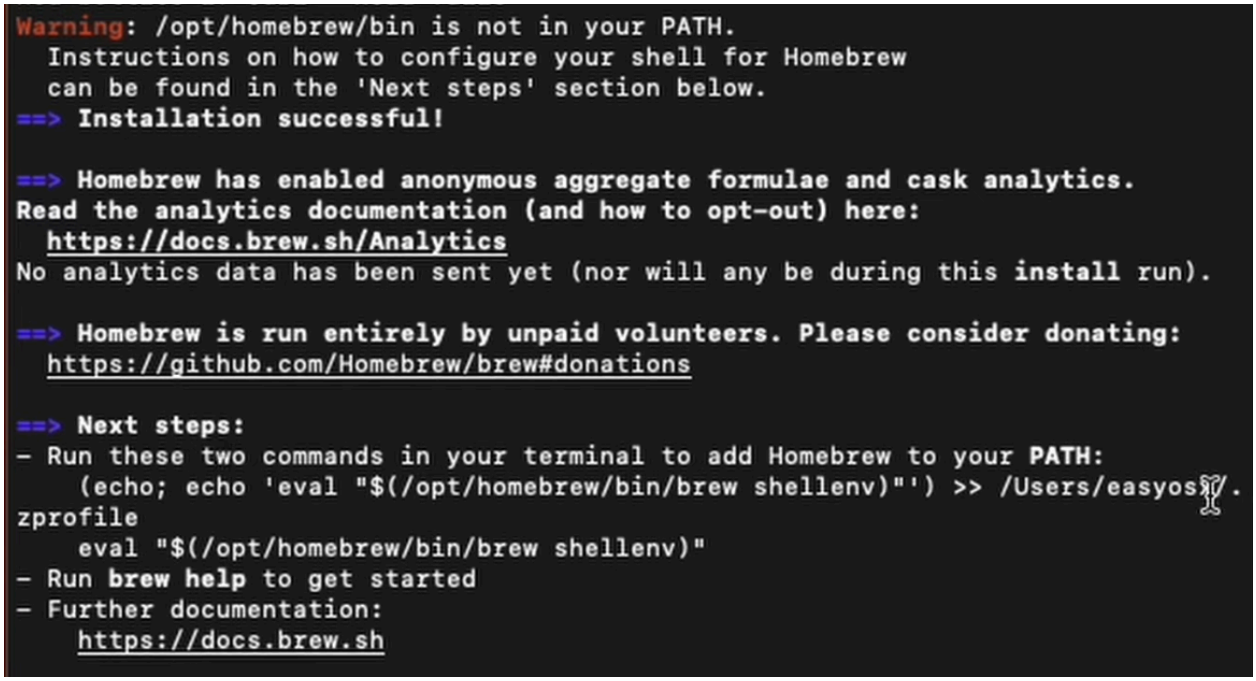
Copy this into your terminal:
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"



From here, you can press the "return" key and may be prompted to enter the password to your computer. Note: Your password may be hidden so just type it in and press return. Afterwards, you should get a message like this:



Press return when you're ready to continue.

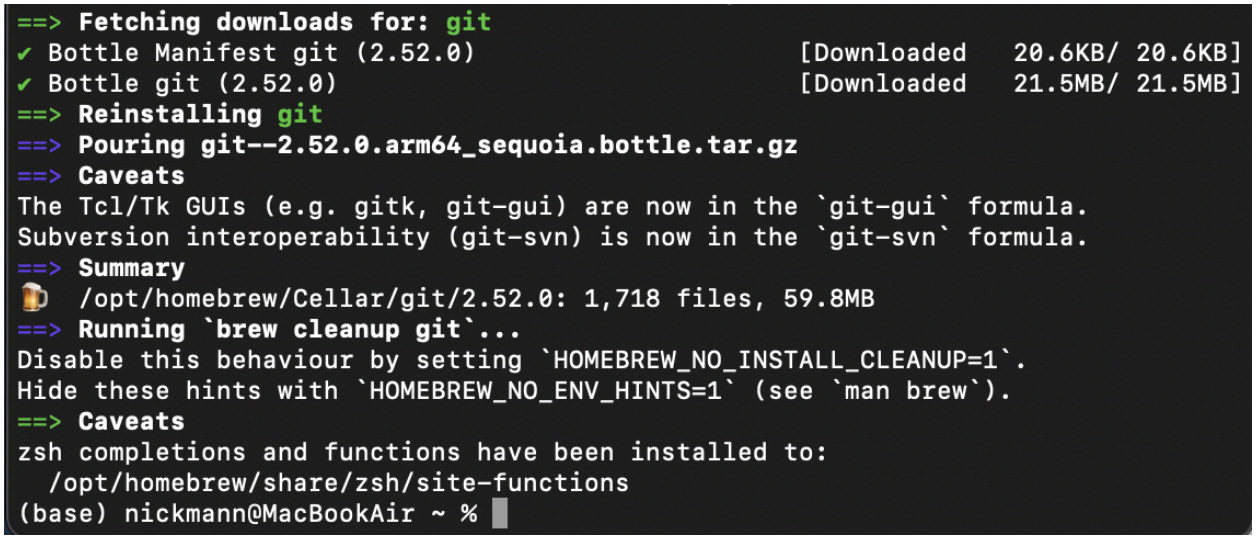


If that runs successfully, you should see this message above (or a similar message).
The "Next steps" section gives you two (*or three) lines of code to run in your terminal. Copy the first one into the terminal, then click return. Then scroll up, copy the next one, and click return. Same for the last one.

This should complete your installation of homebrew.

Now, to install Git, you'll want to type in the terminal: "brew install git"
If this runs properly, you should see lots of text verifying the downloads:

(Mine might look slightly different because I've already installed it)



To verify that it worked, type the following into your terminal: git --version
Another thing you should install is libgit2, which you can do easily by typing in your terminal "brew install libgit2"
This will help you connect GitHub to R later
Now, proceed to the next step (setting up GitHub)

Windows - Installing Git:

Go to the Git website https://git-scm.com/downloads/win

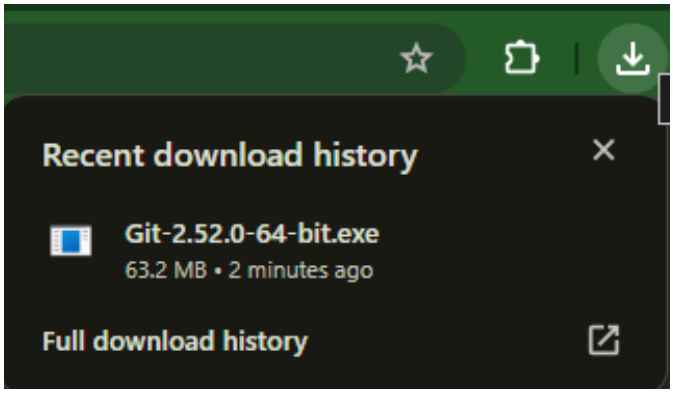
Select your operating system



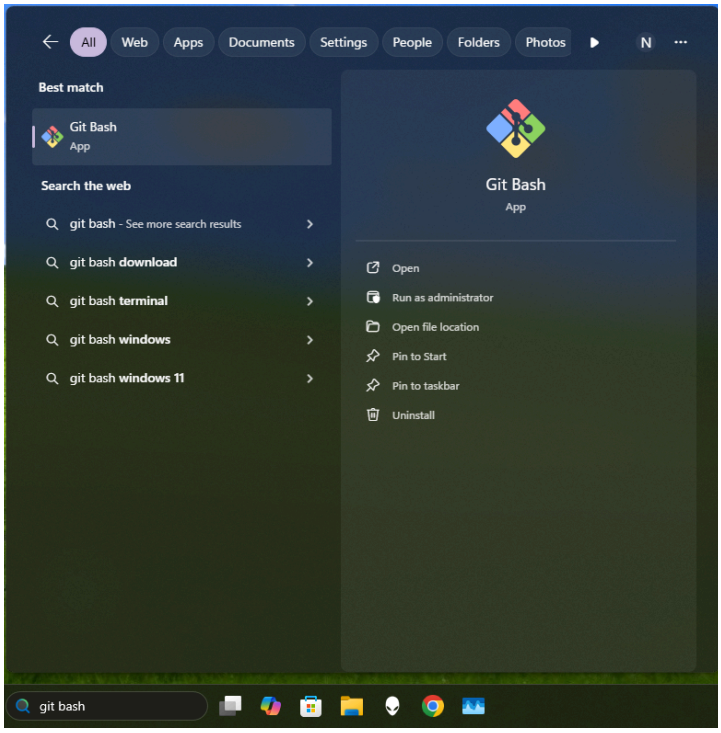
Click on the first line below this section that says “Click here to download”



When the download finishes, run the .exe file.



After reading the terms and conditions, click “Next”.
Select the folder where you would like to install Git, then click “Next”.
Click “Next” to allow git to create a Start Menu folder, and click “Next” again to select the default text editor.
From here, you can click through the remaining default options, reading through if you want more details.
A window will appear titled “Completing the Git Setup Wizard”. Click “Finish”
In your computer search bar on the bottom of your screen (not your browser), type “Git Bash” and click on the app.



Once it opens, type “git –version” and press the “Enter” key



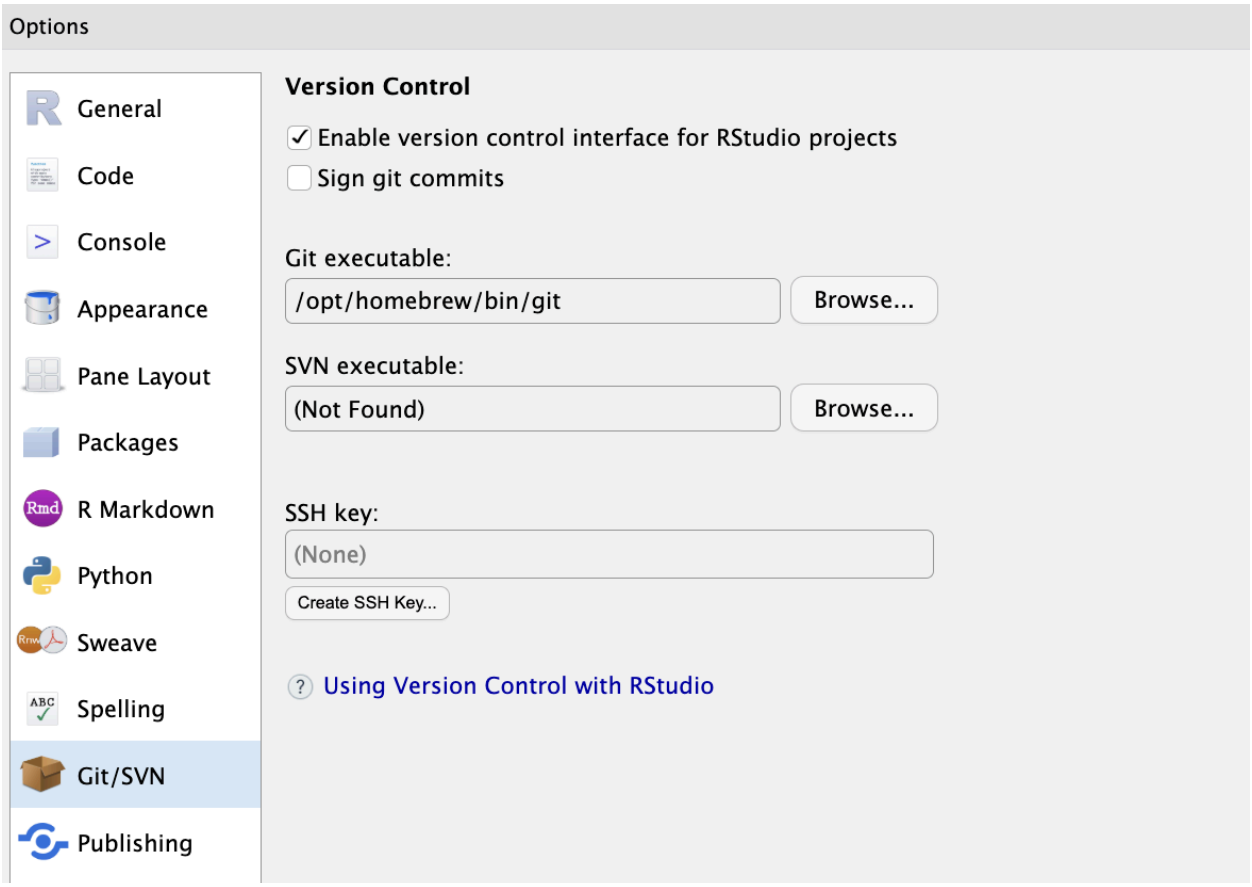
If your installation was successful, you’ll see text verifying the version of Git that you just installed:



Now, proceed to the next step (setting up GitHub)

Setting up GitHub:

How to create a repository in RStudio
Open RStudio, click on “Tools”, then click “Global Options”. Next, select the “Git/SVN” option.



From here, you’ll want to click “Create SSH Key”. Creating a passkey is optional. You can copy the key by pressing “View public key”. Now, go to github.com, click on your profile, then settings, then “SSH and GPG keys”.
Click the green button titled “New SSH key”, title your Key, and copy your SSH key to the text box where it says “Key”. You may be asked to enter your Github password here (you can go ahead and do it).

*Install the “devtools” and “usethis” packages by typing the following lines into the console:
install.packages("devtools")
install.packages("usethis")

After making sure you’ve installed the “devtools” and “usethis” packages, type the following lines of code in the console in RStudio filling in your name and the email address associated with your GitHub account.
Note: Your name is the name in your GitHub profile.

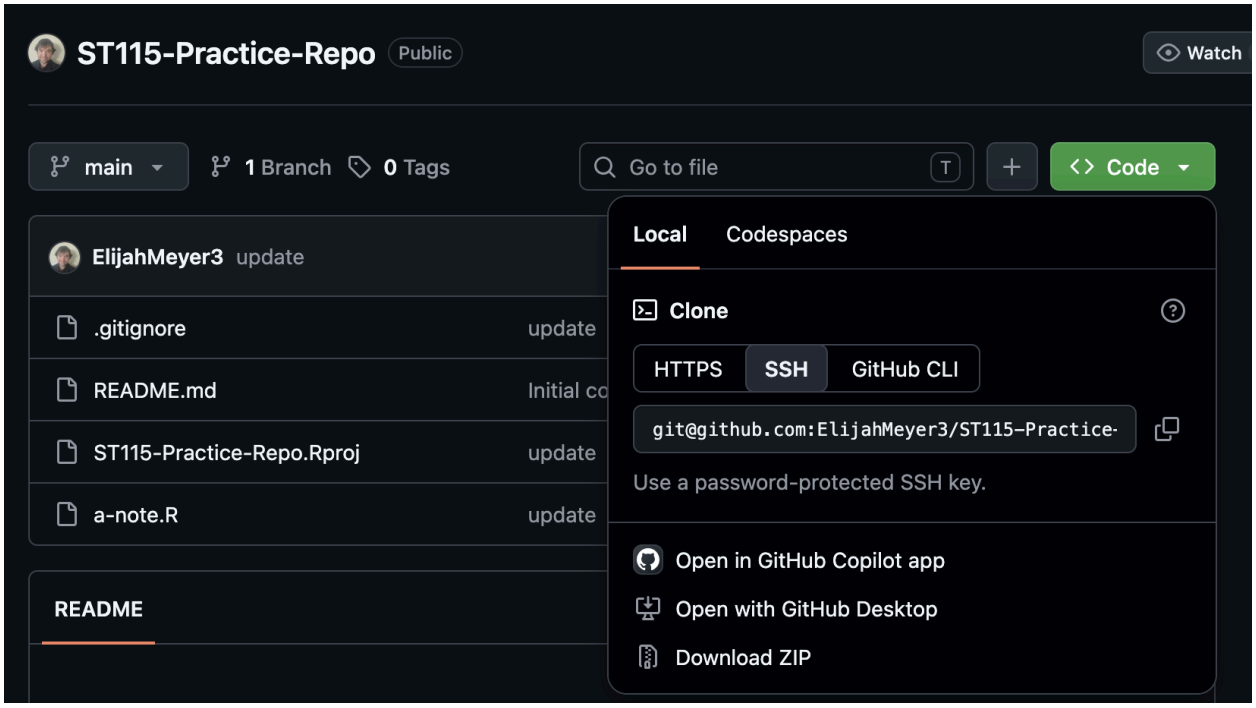
```
devtools::install_github("r-lib/gert")

usethis::use_git_config(
  user.name = "Your name",
  user.email = "Email associated with your GitHub account"
  )
```

Cloning a Repo:

Follow this link to find a practice repository
https://github.com/ElijahMeyer3/ST115-Practice-Repo

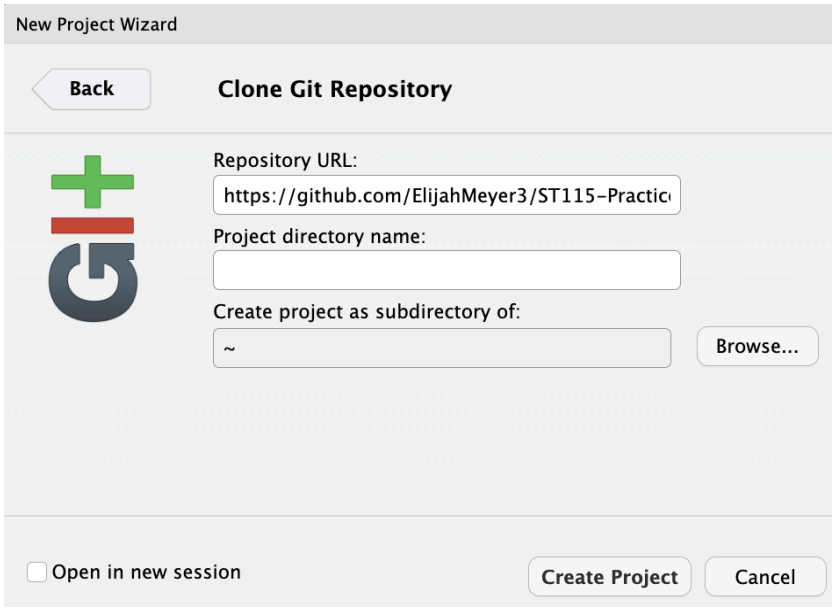
Click on the green “Code” button, click “SSH” below it, and you will see the following.



There is a link in the tab we opened under “SSH”. Copy it using the button to its right. We will use it later.

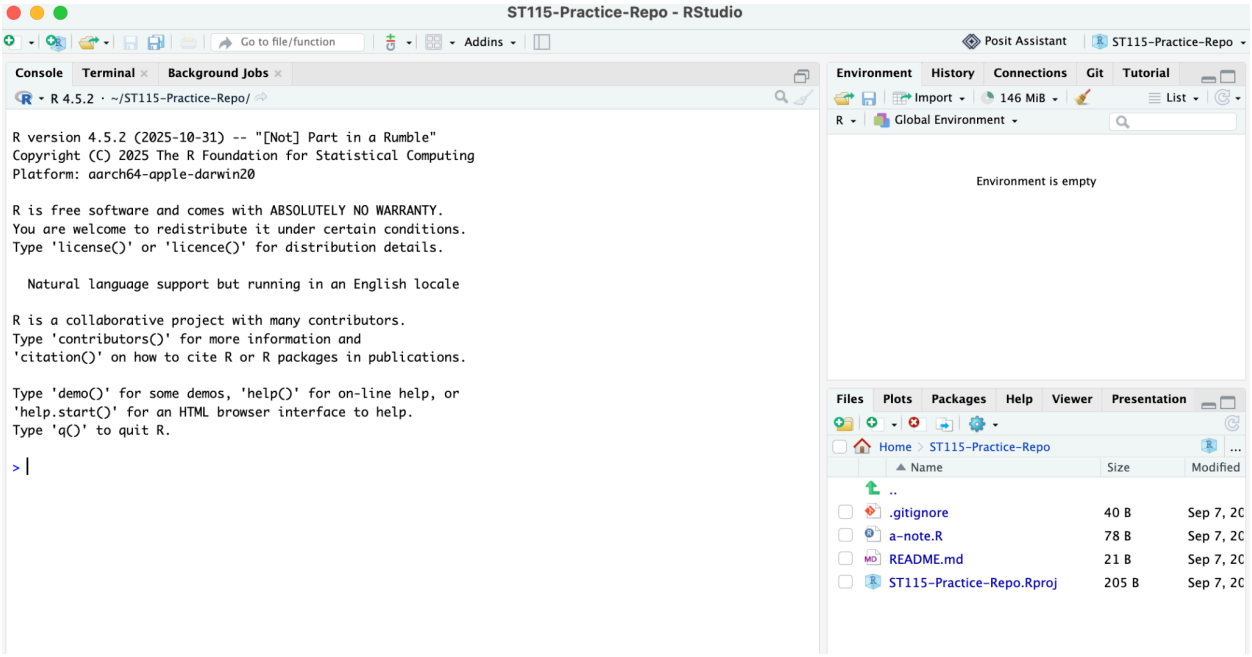
In RStudio, go to “File” then “New Project”. You can save your current workspace if prompted. Select “Version Control” then “Git”

A window will appear titled “Clone Git Repository”. It will ask for a URL. This is where you can paste the link you copied from GitHub earlier. After pasting, it should look like this:



Do not worry about naming the project. If you click “Create project as subdirectory of:” you can select where it will be stored locally on your computer. This is up to you, just make sure you will be able to find it.

Now click “Create Project”. You can verify if this worked by seeing your “Files” tab in the bottom right side of RStudio. It should look like this.



We see the same files that were on GitHub in our “Files” tab.

If this worked, you should be all set!

Tip: In the future, you can open an existing project by clicking “File” then “Open Project”.